MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 2 Solutions

Ex. 2.1.22, 24 Determine which of the pairs of statement forms are logically equivalent. Justify your answers using truth tables and include a few words of explanation.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$.

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$.

SOLUTION . **22.**

p	q	r	$q \vee r$	$p \wedge q$	$p \wedge r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
1	1	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

same truth values

The truth table shows that $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ always have the same truth values. Therefore they are logically equivalent. This proves the distributive law for $\wedge$ over $\vee$.

24.

p	q	r	$p \vee q$	$p \wedge r$	$(p \vee q) \vee (p \wedge r)$	$(p \vee q) \wedge r$	
1	1	1	1	1	1	1	
1	1	0	1	0	1	0	$\leftarrow$
1	0	1	1	1	1	1	
1	0	0	1	0	1	0	$\leftarrow$
0	1	1	1	0	1	1	
0	1	0	1	0	1	0	$\leftarrow$
0	0	1	0	0	0	0	
0	0	0	0	0	0	0	

different truth values

The truth table shows that $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$ have different truth values in rows 2, 4, and 6. Hence they are not logically equivalent. $\square$

Ex. 2.1.28. Use De Morgan's laws to write a negation for the statement

"This computer program has a logical error in the first ten lines or it is being run with an incomplete data set."

SOLUTION . This computer program does not have a logical error in the first ten lines and it is not being run with an incomplete data set.

Note the statement "This computer program does not have a logical error in the first ten lines and it is being run with a complete data set." is not an acceptable answer. The negation of "it is being run with an incomplete data set" is not "it is being run with a complete data set". For example, it could be the case that the program is not running at all. □

Ex. 2.1.33. Assume x is a particular real number and use De Morgan's laws to write negations for the statement

$$-10 < x < 2.$$

SOLUTION .

$$-10 \geq x \quad \text{or} \quad x \geq 2.$$

(Here we are assuming $x \in \mathbb{R}$.) □

Ex. 2.1.42. Use a truth table to establish if the statement form is a tautology or a contradiction.

$$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q.$$

SOLUTION . The truth table can be done like this:

p	q	r	$\neg p$	$\neg p \wedge q$	$q \wedge r$	$\neg q$	$(\neg p \wedge q) \wedge (q \wedge r)$	$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$
1	1	1	0	0	1	0	0	0
1	1	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0
1	0	0	0	0	0	1	0	0
0	1	1	1	1	1	0	1	0
0	1	0	1	1	0	0	0	0
0	0	1	1	0	0	1	0	0
0	0	0	1	0	0	1	0	0

all 0's

Since all the truth values of $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ are 0's, $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ is a contradiction. □

Ex. 2.1.49. Supply a reason for each step.

$$\begin{aligned}(p \vee \neg q) \wedge (\neg p \vee \neg q) &\equiv (\neg q \vee p) \wedge (\neg q \vee \neg p) && \text{by } \underline{\text{(a)}}\\ &\equiv \neg q \vee (p \wedge \neg p) && \text{by } \underline{\text{(b)}}\\ &\equiv \neg q \vee \mathbf{F} && \text{by } \underline{\text{(c)}}\\ &\equiv \neg q && \text{by } \underline{\text{(d)}}\end{aligned}$$

Therefore, $(p \vee \neg q) \wedge (\neg p \vee \neg q) \equiv \neg q$.

SOLUTION .

$$\begin{aligned}(p \vee \neg q) \wedge (\neg p \vee \neg q) &\equiv (\neg q \vee p) \wedge (\neg q \vee \neg p) && \text{by } \underline{\text{commutative law}}\\ &\equiv \neg q \vee (p \wedge \neg p) && \text{by } \underline{\text{distributive law}}\\ &\equiv \neg q \ \vee \mathbf{F} && \text{by } \underline{\text{negation law}}\\ &\equiv \neg q && \text{by } \underline{\text{identity law}}.\end{aligned}$$

□

Ex. 2.1.52. Use the logical laws to verify the logical equivalence

$$\neg(p \vee \neg q) \vee (\neg p \wedge \neg q) \ \equiv \ \neg p.$$

Supply a reason for each step.

SOLUTION .

$$\begin{aligned}\neg(p \vee \neg q) \vee (\neg p \wedge \neg q) &\equiv (\neg p \wedge \neg(\neg q)) \vee (\neg p \wedge \neg q) && \text{by De Morgan's law}\\ &\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) && \text{by double negation}\\ &\equiv \neg p \wedge (q \vee \neg q) && \text{by distributive law}\\ &\equiv \neg p \ \wedge \mathbf{T} && \text{by negation law}\\ &\equiv \neg p && \text{by identity law.}\end{aligned}$$

□

Ex. 2.2.12 (Modified). Apply Ex. 2.2.13a (below) to establish the logical equivalence

$$p \vee q \to r \ \equiv \ (p \to r) \wedge (q \to r).$$

Next, use it rewrite the following statement. (Assume that x represents a fixed real number.)

If $x > 2$ or $x < -2$, then $x^2 > 4$.

SOLUTION . (a)

$$\begin{aligned}p \vee q \to r &\equiv \neg(p \vee q) \vee r && \text{by Ex. 2.2.13a}\\ &\equiv (\neg p \wedge \neg q) \vee r && \text{by De Morgan's law}\\ &\equiv r \vee (\neg p \wedge \neg q) && \text{by commutativity law}\\ &\equiv (r \vee \neg p) \wedge (r \vee \neg q) && \text{by distributive law}\\ &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{by commutativity law}\\ &\equiv (p \to r) \wedge (q \to r) && \text{by Ex. 2.2.13a}\end{aligned}$$

(b) If $x > 2$ then $x^2 > 4$, and if $x < -2$ then $x^2 > 4$. □

Ex. 2.2.13a. Use a truth table to verify the following logical equivalence. Include a few words of explanation with your answer.

$$p \to q \ \equiv\ \neg p \vee q.$$

SOLUTION .

p	q	$p \to q$	$\neg p \vee q$
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	1

same truth values

The table shows that $p \to q$ and $\neg p \vee q$ have the same truth values. Therefore, they are logically equivalent. □

Question E1. Are the following statement forms tautologies, contradictions, or neither?

a. $p \wedge (p \vee \neg p)$
b. $\neg(p \vee \neg p) \vee p$
c. $\neg(p \wedge q) \vee p$

SOLUTION . a. $p \wedge (p \vee \neg p) \equiv p \wedge \mathbf{T} \equiv p$. (We used negation law and identity law). So neither tautology nor contradiction.

b. $\neg(p \vee \neg p) \vee p \equiv \neg\mathbf{T} \vee p \equiv \mathbf{F} \vee p \equiv p$. (We used negation law and identity law). So neither tautology nor contradiction.

c.

$$\begin{aligned}
\neg(p \wedge q) \vee p &\equiv (\neg p \vee \neg q) \vee p && \text{(De Morgan's Law)} \\
&\equiv (\neg q \vee \neg p) \vee p && \text{(Commutative Law)} \\
&\equiv \neg q \vee (\neg p \vee p) && \text{(Associative Law)} \\
&\equiv \neg q \vee \mathbf{T} && \text{(Negation Law)} \\
&\equiv \mathbf{T} && \text{(Universal Bound Law)}
\end{aligned}$$

It is a tautology.

□